

Automated Essay Scoring in Education

Jing Yang
School of Computer Science
Central China Normal University
Wuhan, China
yangj987@mails.ccn.edu.cn

Abstract—As artificial intelligence advances rapidly, Automated Essay Scoring (AES) technology has made significant strides, effectively addressing the challenges of high labor intensity and slow feedback encountered by teachers during essay grading. In recent years, numerous innovative research findings and methodologies have emerged in this field. We explore in this paper the mainstream studies in AES and categorize them into two groups: single-prompt AES and cross-prompt AES. Single-prompt AES involves scoring scenarios where training and test essays are written for the same prompt, comprising both feature engineering and deep learning-based approaches. On the other hand, cross-prompt AES entails scoring scenarios where training and test essays are written for different prompts, often implemented within the frameworks of contrast learning, transfer learning, or multi-task learning. Drawing from existing research, we analyze the primary issues and challenges in AES and proposes potential directions for future research.

Keywords—Automated essay scoring, BERT, Neural network, Artificial intelligence

I. INTRODUCTION

Essays play a crucial role in language learning. They allow students to organize thoughts, express themselves, and showcase creativity. Additionally, essays are an important tool for assessing students' writing levels. At a time when the number of students is growing and online education is becoming increasingly important, automated grading systems are critical. It efficiently processes large volumes of student essay data and provides in-depth data analysis for education administrators and policymakers. Teachers often encounter various challenges in traditional manual scoring processes, including scoring fatigue, personal preferences, and biases, adopt AES systems can significantly enhance fairness, consistency, and efficiency in the scoring process. In recent years, with the advancement of artificial intelligence, AES has overcome the drawbacks of manual scoring to some extent, liberating teachers from repetitive scoring tasks. However, AES systems inevitably have some limitations. Exploring the advantages and limitations of artificial intelligence in the AES field lays the foundation for researching effective and accurate AES processes. Based on the current research status of AES systems, this paper aims to elucidate the advantages and disadvantages of various methods in the AES field and explore the challenges and possible trends of AES system development by integrating the latest machine learning technology and the requirements of the AES field.

II. SINGLE-PROMPT AES

A. AES Based on Feature Engineering

Conventional AES systems typically employ feature engineering, encompassing regression, classification, and ranking as their primary methodologies, which rely on manually crafted features, such as word count, grammatical errors, lexical complexity, syntactic structures, and readability metrics. These features are meticulously designed by researchers based on linguistic expertise, allowing for adaptability to various scoring criteria. Despite the reliance on hand-crafted features, which are exceptional accuracy, conventional AES systems demonstrate robust performance even with limited datasets.

AES is often considered as a regression problem due to its focus on evaluating the essays quality. For instance, the early PEG system utilized multiple linear regression techniques to construct scoring models incorporating specific feature variables. Attali and Burstein [1] assessed the widely adopted E-Rater system, which employs linear regression to identify crucial features for predicting essay scores. Both systems predominantly employ linear models for score prediction, indicating an initial emphasis on feature selection in AES development. Phandi et al. [2] leveraged EASE to extract features and differentiate between high and low-scoring essays, employing BLRR as a learning algorithm to enhance AES effectiveness through domain adaptation, highlighting the importance of model generalization. Over time, AES has evolved from solely scoring essays to conducting more intricate text analyses. Persing et al. [3], [4] pioneered the assessment of organizational structure and argument strength in essays by designing tailored features for these tasks, thereby enhancing AES performance. Zesch et al. [5] investigated feature transferability and interdependence across different AES tasks, introducing readability features. Cozma et al. [6] proposed an AES approach combining string kernels and word embeddings, employing v-Support Vector Regression for model training.

AES is commonly seen as a task of text classification, wherein various text characteristics like grammatical complexity, lexical complexity, lexical chain, coherence, etc., are utilized for classification purposes. The model directly predicts labels corresponding to different scores based on these features. Early studies in this domain, as conducted by Larkey [7] and Rudner and Liang [8], employed artificial feature extraction coupled with a Bayesian classifier for AES. Vajjala [9] employed a sequential minimal optimization algorithm for training the

classification model. However, this classification approach cannot directly yield essay scores. It could potentially be integrated with deep learning or regression methods in the future to enhance the accuracy of score predictions.

AES can be viewed as a text feature sorting task, encompassing elements like n-grams, syntactic complexity, and parse tree features. Yannakoudakis et al. [10] employed SVMRank for score ranking, leveraging a combination of lexical, discourse and syntactic features automatically extracted from the text to model hierarchical relationships between essays. Chen and He [11] introduced the tabular learning ranking method, which integrates human and machine rater agreement directly into the loss function to enhance AES accuracy. However, this sorting approach does not yield direct scores and demands extensive data and computational resources. In the future, combining sorting methods with deep learning techniques could potentially enhance sorting accuracy.

B. AES Based on Traditional Deep Learning Models

Feature engineering-based AES faces numerous limitations. Manual feature extraction is both time-consuming and labor-intensive, yielding only shallow features. Given the significant strides made by deep learning in natural language processing, its integration into AES tasks emerges as a promising avenue for research. This approach involves transforming text into low-dimensional vector representations, extracting text features, and ultimately predicting final scores. The subsequent sections will delve into traditional deep learning models applied in AES in detail.

In AES tasks, comprehending the entire essay's structure and content is essential. RNN are gaining attention due to their cyclic structure, which enables them to process information flow in text, allowing for the understanding of temporal and contextual relationships. Researchers are increasingly intrigued by the potential of RNNs in AES. For instance, Taghipour and Ng [12] explored various neural network methods, including RNNs, and discovered that LSTM outperforms traditional RNNs and GRUs in processing textual representations for AES tasks. Alikaniotis et al. [13] compared LSTM with BiLSTM, showing that BiLSTM can effectively manage both forward and backward dependencies in text, offering a more comprehensive textual representation. Wang et al. [14] utilized BiLSTM to represent text and introduced Dilated LSTM to capture sequence information at different levels by adjusting the skip connections. Muangkammuen and Fukumoto [15] employed BiLSTM at both word and sentence levels, integrating attention mechanisms with multitasking learning for enhanced feature robustness. These initial investigations into RNNs and their variants, such as LSTM and BiLSTM, underscore their benefits in processing text sequences for AES. Convolutional Neural Networks (CNNs) have also garnered attention for their capacity to capture local text features like lexical usage and grammatical structure. Taghipour and Ng [12] leveraged convolutional layers to extract local features, enhancing the model's ability to capture and process textual information locally. With advancements in technology, researchers have begun exploring more intricate structures. Dong and Zhang [16] developed a hierarchical CNN model, representing sentence structures at lower levels and leveraging them to express the entire essay's structure at upper

levels. This structure enhances the model's comprehension and assessment of both the organizational structure and content quality within the article. Dong et al. [17] proposed the RCNN model, amalgamating CNN's proficiency in handling local features with RNN's prowess in managing sequence data. Through the incorporation of an attention mechanism, the RCNN model is further refined to capture pivotal information effectively. Furthermore, Mathias et al. [18] integrated LSTM with CNN within a multi-task learning framework to augment the accuracy and resilience of automated essay scoring. Tay et al. [19] introduced the SKIPFLOW mechanism, leveraging the neural tensor layer to grasp semantic relationships among hidden LSTM states across different time points. Building upon this, Tay et al. [20] extended their research by integrating the SKIPFLOW mechanism with the foundational LSTM model, effectively amalgamating logical and semantic text flows, thus enhancing the model's processing capabilities.

Research indicates that certain well-crafted adversarial inputs can attain high scores within AES systems, prompting concerns regarding the model's ability to discern adversarial samples. Farag et al. [21] proposed the LC model, designed to assess and enhance text coherence at a local level. Liu et al. [22] introduced the TSLF framework, which merges advantages from both feature engineering and end-to-end models. They concatenated three scores obtained from the neural network with manual features at the outset, feeding these combined features into the XGBoost model for additional training, thereby bolstering the model's resilience against adversarial samples.

Conventional deep learning models in essay scoring primarily aim to improve the interpretability of their results. Researchers employ trait-specific grading to gain deeper insights into the essay quality. Chae and Nenkova [23] use shallow structural features and multilayer perceptron classifiers to predict text fluency, distinguishing between machine and human translation fluency. Persing et al. develop various models for evaluating different writing traits, including organization [24] and thesis clarity [25]. They also examine prompt adherence [26], argument strength [27], and stance scoring models [28]. Somasundaran et al. [29] propose a feature-rich method for scoring essays on Prompt Adherence. Taghipour [30] scores organization and argument strength, while Mathias et al. [31] score organization, coherence, and cohesion. Mathias and Bhattacharyya [32] evaluate writing styles based on goodness, and Somasundaran et al. [33] assess narrative quality. Ke et al. [34] study thesis articulation. Kumar et al. [35] propose a multi-task learning framework where the overall essay score is the primary task, and individual essay trait scores serve as auxiliary tasks for comprehensive evaluation.

C. AES Based on Pre-trained Models

Lately, the impressive performance of large pre-trained language models, like BERT, has spurred researchers to explore their application in AES tasks for enhanced results. BERT, a Transformer-based model, enhances text representation by training bidirectionally, taking into account both preceding and succeeding words in context. Its methodology involves initial pre-training on extensive textual data to grasp intricate language features, followed by swift enhancement of task-specific performance through fine-tuning. On the other hand, XLNet

integrates autoregressive language modeling advantages with self-encoder strengths, aiming to refine model performance and versatility via a more intricate training approach.

Rodriguez et al. [36] and Uto et al. [37] utilize BERT and XLNet for scoring purposes. Mayfield and Black [38] delve into refining the application of BERT's pre-training model within the AES domain, employing a periodic learning rate strategy for fine-tuning. Yang et al. [39] introduce R²BERT, a novel model achieved through fine-tuning BERT and integrating regression and ranking losses. Wang et al. [40] leverage multiple loss functions and transfer learning alongside pre-trained BERT models, primarily focusing on fine-tuning. Cao et al. [41] devise a domain adaptation framework to enhance model robustness against domain shifts by introducing perturbations. Song et al. [42] assess text organizational structure by incorporating discourse elements, utilizing a pre-training method to grasp diverse language representations effectively. Xie et al. [43] propose the NPCR model, employing contrastive learning principles, BERT for feature extraction, and optimizing both regression and ranking objectives.

D. Performance

The performance of some classic AES models is listed in Table I. Single-prompt AES models have matured significantly in recent years, especially with pre-trained models achieving unprecedented levels of accuracy. This advancement highlights the growing potential and effectiveness of automated essay scoring systems in educational contexts.

TABEL I. QWK EXPERIMENTAL RESULTS OF EACH MODEL IN ASAP

Type	Model	Avg
Feature engineering	EASE(SVR) [2]	0.699
	EASE(BLRR) [2]	0.705
deep learning	CNN [16]	0.726
	RNN [12]	0.675
	LSTM [13]	0.756
	SKIPFLOW [19]	0.764
	R ² BERT [39]	0.794
pre-trained	NPCR [43]	0.817

III. CROSS-PROMPT AES

The AES models discussed earlier typically require training for specific prompts, yet practical application scenarios often lack sufficient training data. This scarcity poses challenges in applying existing single-prompt AES models, prompting a shift in focus towards cross-prompt AES. Cozma et al. [6] and Dong and Zhang [16] conducted experiments in both cross-domain and in-domain conditions, revealing suboptimal cross-prompt effects and highlighting the unsuitability of the single-prompt method for cross-prompt AES. Consequently, researchers are delving into methodologies tailored specifically for cross-prompt AES.

A. AES with Contrastive Learning

Chen and Li [44] introduced PMAES, employing mapping contrast learning to improve the consistency of across-prompts AES systems. Jiang et al. [45] devised the prompt perception neural AES model, effectively distinguishing between the quality and compliance features of essays through a comparison

of the norm-angle alignment strategy and the counterfactual self-training strategy.

B. AES with Transfer Learning

Phandi et al. [2] introduced a domain-adaptive AES technique utilizing Bayesian linear ridge regression. This method can adjust to new prompts by incorporating manual features and a small dataset of new prompt data. Jin et al. [46] pioneered the TDNN two-stage model, marking a significant step in addressing cross-prompt AES challenges. Building upon the TDNN framework, Li et al. [47] developed the SEDNN model. The model combines knowledge from multiple source prompts in two stages to identify features that can be transferred to target prompts. Liu and Ding [48] proposed the MFDNN model, expanding upon the TDNN and SEDNN concepts with a three-stage architecture - a pre-scoring model, a dependency information extraction model and a composite scoring model. Additionally, Ridley et al. [49] presented PAES, a rating system independent of specific cue data, leveraging parts-of-speech embedding and multidimensional features for scoring.

C. AES with Multitasking Learning

Cummins et al. [50] introduced task-specific variables into pin-preference learning techniques to facilitate cross-task learning. Ridley et al. [51] achieved precise prediction of article feature scores using the CTS multi-task model and a feature attention mechanism, building upon previous methods. Ding et al. [52] enhanced scoring accuracy by integrating a multi-task framework for automated tasks. Meanwhile, Do et al. [53] proposed the ProTACT model, leveraging attention mechanisms and LDA to optimize scoring and enhance system performance. These studies collectively contribute to enhancing the accuracy of AES systems through multi-task learning methodologies.

D. Performance

Table II provides experimental results of cross-prompt AES models, which generally exhibit lower average QWK scores compared to established single-prompt AES models. This indicates ongoing challenges in achieving comparable performance across different essay prompts, emphasizing the need for further improvement and development in cross-prompt AES methodologies.

TABEL II. QWK EXPERIMENTAL RESULTS OF EACH MODEL IN ASAP

Type	Model	Avg
contrastive	PMAES [44]	0.566
transfer	TDNN [46]	0.725
	PAES [49]	0.686
multitasking	ProTACT [53]	0.592

IV. CONCLUSIONS AND FUTURE WORK

The AES system aims to streamline the essay scoring process by automatically evaluating essay quality through artificial intelligence algorithms, thereby reducing the time and resources required for manual scoring. While AES offers rapid feedback and alleviates the scorer burden, it faces several inherent limitations. Firstly, AES systems heavily rely on the quality and scope of training data, constraining their ability to assess specialized or domain-specific essays. Additionally, the diverse array of assessment metrics employed in AES studies,

such as accuracy, F1 score, relevance, or QWK, complicates cross-study comparisons and lacks standardized evaluation criteria. Secondly, most AES systems prioritize grammatical and structural accuracy over content depth and logical coherence. Consequently, AES may inadvertently favor verbose yet superficial and incoherent essays, lacking the capacity to offer nuanced feedback or personalized improvement suggestions.

In conclusion, while AES systems have the potential to improve scoring efficiency, they are far from mature in their ability to understand advanced writing skills and depth of content. Future AES research needs to focus on how to improve the domain adaptability, assessment accuracy, and feedback quality of the systems to better serve educational purposes and truly aid in the educational assessment process rather than simply replace human raters.

REFERENCES

- [1] Y. Attali and J. Burstein, "Automated Essay Scoring With e-rater® V.2", *JTLA*, vol. 4, no. 3, Feb. 2006.
- [2] Peter Phandi, Kian Ming A. Chai, and Hwee Tou Ng. 2015. Flexible Domain Adaptation for Automated Essay Scoring Using Correlated Linear Regression. In *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing*, pages 431–439, Lisbon, Portugal. Association for Computational Linguistics.
- [3] Isaac Persing, Alan Davis, and Vincent Ng. 2010. Modeling Organization in Student Essays. In *Proceedings of the 2010 Conference on Empirical Methods in Natural Language Processing*, pages 229–239, Cambridge, MA. Association for Computational Linguistics.
- [4] Isaac Persing and Vincent Ng. 2015. Modeling Argument Strength in Student Essays. In *Proceedings of the 53rd Annual Meeting of the Association for Computational Linguistics and the 7th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*, pages 543–552, Beijing, China. Association for Computational Linguistics.
- [5] Torsten Zesch, Michael Wojatzki, and Dirk Scholten-Akoun. 2015. Task-Independent Features for Automated Essay Grading. In *Proceedings of the Tenth Workshop on Innovative Use of NLP for Building Educational Applications*, pages 224–232, Denver, Colorado. Association for Computational Linguistics.
- [6] Mădălina Cozma, Andrei Butnaru, and Radu Tudor Ionescu. 2018. Automated essay scoring with string kernels and word embeddings. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 503–509, Melbourne, Australia. Association for Computational Linguistics.
- [7] Larkey, & Leah, S. . (2019). Automatic essay grading using text categorization techniques.
- [8] L. M. Rudner and T. Liang, "Automated Essay Scoring Using Bayes' Theorem", *JTLA*, vol. 1, no. 2, Jun. 2002.
- [9] Vajjala, S. (2016). Automated Assessment of Non-Native Learner Essays: Investigating the Role of Linguistic Features. *International Journal of Artificial Intelligence in Education*, 28, 79 - 105.
- [10] Helen Yannakoudakis, Ted Briscoe, and Ben Medlock. 2011. A New Dataset and Method for Automatically Grading ESOL Texts. In *Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics: Human Language Technologies*, pages 180–189, Portland, Oregon, USA. Association for Computational Linguistics.
- [11] Hongbo Chen and Ben He. 2013. Automated Essay Scoring by Maximizing Human-Machine Agreement. In *Proceedings of the 2013 Conference on Empirical Methods in Natural Language Processing*, pages 1741–1752, Seattle, Washington, USA. Association for Computational Linguistics.
- [12] Kaveh Taghipour and Hwee Tou Ng. 2016. A Neural Approach to Automated Essay Scoring. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, pages 1882–1891, Austin, Texas. Association for Computational Linguistics.
- [13] Dimitrios Alikanotis, Helen Yannakoudakis, and Marek Rei. 2016. Automatic Text Scoring Using Neural Networks. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 715–725, Berlin, Germany. Association for Computational Linguistics.
- [14] Yucheng Wang, Zhongyu Wei, Yaqian Zhou, and Xuanjing Huang. 2018. Automatic Essay Scoring Incorporating Rating Schema via Reinforcement Learning. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing*, pages 791–797, Brussels, Belgium. Association for Computational Linguistics.
- [15] Panitan Muangkammuen and Fumiyo Fukumoto. 2020. Multi-task Learning for Automated Essay Scoring with Sentiment Analysis. In *Proceedings of the 1st Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 10th International Joint Conference on Natural Language Processing: Student Research Workshop*, pages 116–123, Suzhou, China. Association for Computational Linguistics.
- [16] Fei Dong and Yue Zhang. 2016. Automatic Features for Essay Scoring – An Empirical Study. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing*, pages 1072–1077, Austin, Texas. Association for Computational Linguistics.
- [17] Fei Dong, Yue Zhang, and Jie Yang. 2017. Attention-based Recurrent Convolutional Neural Network for Automatic Essay Scoring. In *Proceedings of the 21st Conference on Computational Natural Language Learning (CoNLL 2017)*, pages 153–162, Vancouver, Canada. Association for Computational Linguistics.
- [18] Sandeep Mathias, Rudra Murthy, Diptesh Kanojia, Abhijit Mishra, and Pushpak Bhattacharyya. 2020. Happy Are Those Who Grade without Seeing: A Multi-Task Learning Approach to Grade Essays Using Gaze Behaviour. In *Proceedings of the 1st Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics and the 10th International Joint Conference on Natural Language Processing*, pages 858–872, Suzhou, China. Association for Computational Linguistics.
- [19] Yi Tay, Minh C Phan, Luu Anh Tuan, and Siu Cheung Hui. 2017. Skipflow: Incorporating neural coherence features for end-to-end automatic text scoring. *arXiv preprint arXiv:1711.04981*.
- [20] Yi Tay, Minh C. Phan, Luu Anh Tuan, and Siu Cheung Hui. 2018. Skipflow:incorporating neural coherence features for end-to-end automatic text scoring. In *Proceedings of the Thirty-Second AAAI Conference on Artificial Intelligence*, pages 5948–5955.
- [21] Youmna Farag, Helen Yannakoudakis, and Ted Briscoe. 2018. Neural Automated Essay Scoring and Coherence Modeling for Adversarially Crafted Input. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long Papers)*, pages 263–271, New Orleans, Louisiana. Association for Computational Linguistics.
- [22] Liu, J., Xu, Y., & Zhao, L. (2019). Automated Essay Scoring based on Two-Stage Learning. *ArXiv*, abs/1901.07744.
- [23] Chae, J., & Nenkova, A. (2009). Predicting the Fluency of Text with Shallow Structural Features: Case Studies of Machine Translation and Human-Written Text. *Conference of the European Chapter of the Association for Computational Linguistics*.
- [24] Isaac Persing, Alan Davis, and Vincent Ng. 2010. Modeling organization in student essays. In *Proceedings of the 2010 Conference on Empirical Methods in Natural Language Processing*, pages 229–239, Cambridge, MA. Association for Computational Linguistics.
- [25] Isaac Persing and Vincent Ng. 2013. Modeling Thesis Clarity in Student Essays. In *Proceedings of the 51st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 260–269, Sofia, Bulgaria. Association for Computational Linguistics.
- [26] Swapna Somasundaran, Jill Burstein, and Martin Chodorow. 2014. Lexical Chaining for Measuring Discourse Coherence Quality in Test-taker Essays. In *Proceedings of COLING 2014, the 25th International Conference on Computational Linguistics: Technical Papers*, pages 950–961, Dublin, Ireland. Dublin City University and Association for Computational Linguistics.
- [27] Isaac Persing and Vincent Ng. 2015. Modeling Argument Strength in Student Essays. In *Proceedings of the 53rd Annual Meeting of the Association for Computational Linguistics and the 7th International Joint Conference on Natural Language Processing (Volume 1: Long Papers)*,

- pages 543–552, Beijing, China. Association for Computational Linguistics.
- [28] Isaac Persing and Vincent Ng. 2016. Modeling Stance in Student Essays. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2174–2184, Berlin, Germany. Association for Computational Linguistics.
 - [29] Swapna Somasundaran, Jill Burstein, and Martin Chodorow. 2014. Lexical Chaining for Measuring Discourse Coherence Quality in Test-taker Essays. In *Proceedings of COLING 2014, the 25th International Conference on Computational Linguistics: Technical Papers*, pages 950–961, Dublin, Ireland. Dublin City University and Association for Computational Linguistics.
 - [30] Kaveh Taghipour. 2017. Robust trait-specific essay scoring using neural networks and density estimators. Ph.D. thesis, National University of Singapore (Singapore).
 - [31] Sandeep Mathias, Diptesh Kanojia, Kevin Patel, Samarth Agrawal, Abhijit Mishra, and Pushpak Bhattacharyya. 2018. Eyes are the Windows to the Soul: Predicting the Rating of Text Quality Using Gaze Behaviour. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 2352–2362, Melbourne, Australia. Association for Computational Linguistics.
 - [32] Sandeep Mathias and Pushpak Bhattacharyya. 2018. Thank “Goodness”! A Way to Measure Style in Student Essays. In *Proceedings of the 5th Workshop on Natural Language Processing Techniques for Educational Applications*, pages 35–41, Melbourne, Australia. Association for Computational Linguistics.
 - [33] Swapna Somasundaran, Michael Flor, Martin Chodorow, Hillary Molloy, Binod Gyawali, and Laura McCulla. 2018. Towards Evaluating Narrative Quality In Student Writing. *Transactions of the Association for Computational Linguistics*, 6:91–106.
 - [34] Zixuan Ke, Hrishikesh Inamdar, Hui Lin, and Vincent Ng. 2019. Give Me More Feedback II: Annotating Thesis Strength and Related Attributes in Student Essays. In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 3994–4004, Florence, Italy. Association for Computational Linguistics.
 - [35] Rahul Kumar, Sandeep Mathias, Sripama Saha, and Pushpak Bhattacharyya. 2022. Many Hands Make Light Work: Using Essay Traits to Automatically Score Essays. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 1485–1495, Seattle, United States. Association for Computational Linguistics.
 - [36] Pedro Uria Rodriguez, Amir Jafari, and Christopher M. Ormerod. 2019. Language models and automated essay scoring. In *arXiv: Computation and Language*.
 - [37] Masaki Uto, Yikuan Xie, and Maomi Ueno. 2020. Neural Automated Essay Scoring Incorporating Handcrafted Features. In *Proceedings of the 28th International Conference on Computational Linguistics*, pages 6077–6088, Barcelona, Spain (Online). International Committee on Computational Linguistics.
 - [38] Elijah Mayfield and Alan W Black. 2020. Should You Fine-Tune BERT for Automated Essay Scoring?. In *Proceedings of the Fifteenth Workshop on Innovative Use of NLP for Building Educational Applications*, pages 151–162, Seattle, WA, USA → Online. Association for Computational Linguistics.
 - [39] Ruosong Yang, Jiannong Cao, Zhiyuan Wen, Youzheng Wu, and Xiaodong He. 2020. Enhancing Automated Essay Scoring Performance via Fine-tuning Pre-trained Language Models with Combination of Regression and Ranking. In *Findings of the Association for Computational Linguistics: EMNLP 2020*, pages 1560–1569, Online. Association for Computational Linguistics.
 - [40] Yongjie Wang, Chuang Wang, Ruobing Li, and Hui Lin. 2022. On the Use of Bert for Automated Essay Scoring: Joint Learning of Multi-Scale Essay Representation. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, pages 3416–3425, Seattle, United States. Association for Computational Linguistics.
 - [41] Cao, Y., Jin, H., Wan, X., & Yu, Z. (2020). Domain-Adaptive Neural Automated Essay Scoring. *Proceedings of the 43rd International ACM SIGIR Conference on Research and Development in Information Retrieval*.
 - [42] Wei Song, Ziyao Song, Lizhen Liu, and Ruiji Fu. 2020. Hierarchical multi-task learning for organization evaluation of argumentative student essays. In *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence, IJCAI-20*, pages 3875–3881.
 - [43] Jiayi Xie, Kaiwei Cai, Li Kong, Junsheng Zhou, and Weiguang Qu. 2022. Automated Essay Scoring via Pairwise Contrastive Regression. In *Proceedings of the 29th International Conference on Computational Linguistics*, pages 2724–2733, Gyeongju, Republic of Korea. International Committee on Computational Linguistics.
 - [44] Yuan Chen and Xia Li. 2023. PMAES: Prompt-mapping Contrastive Learning for Cross-prompt Automated Essay Scoring. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1489–1503, Toronto, Canada. Association for Computational Linguistics.
 - [45] Zhiwei Jiang, Tianyi Gao, Yafeng Yin, Meng Liu, Hua Yu, Zifeng Cheng, and Qing Gu. 2023. Improving Domain Generalization for Prompt-Aware Essay Scoring via Disentangled Representation Learning. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 12456–12470, Toronto, Canada. Association for Computational Linguistics.
 - [46] Cancan Jin, Ben He, Kai Hui, and Le Sun. 2018. TDNN: A Two-stage Deep Neural Network for Prompt-independent Automated Essay Scoring. In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1088–1097, Melbourne, Australia. Association for Computational Linguistics.
 - [47] li, X., Chen, M., & Nie, J. (2020). SEDNN: Shared and enhanced deep neural network model for cross-prompt automated essay scoring. *Knowl. Based Syst.*, 210, 106491.
 - [48] Liu, C., & Ding, G. (2021). MFDNN: Mixed Features Deep Neural Network Model for Prompt-independent Automated Essay Scoring. *Proceedings of the 2021 4th International Conference on Algorithms, Computing and Artificial Intelligence*.
 - [49] Ridley, R., He, L., Dai, X., Huang, S., & Chen, J. (2021). Automated Cross-prompt Scoring of Essay Traits. *AAAI Conference on Artificial Intelligence*.
 - [50] Ronan Cummins, Meng Zhang, and Ted Briscoe. 2016. Constrained Multi-Task Learning for Automated Essay Scoring. In *Proceedings of the 54th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 789–799, Berlin, Germany. Association for Computational Linguistics.
 - [51] Ridley, R., He, L., Dai, X.-yu, Huang, S., & Chen, J. (2021). Automated Cross-prompt Scoring of Essay Traits. *Proceedings of the AAAI Conference on Artificial Intelligence*, 35(15), 13745-13753.
 - [52] Yuning Ding, Marie Bexte, and Andrea Horbach. 2023. Score It All Together: A Multi-Task Learning Study on Automatic Scoring of Argumentative Essays. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 13052–13063, Toronto, Canada. Association for Computational Linguistics.
 - [53] Heejin Do, Yunsu Kim, and Gary Geunbae Lee. 2023. Prompt- and Trait Relation-aware Cross-prompt Essay Trait Scoring. In *Findings of the Association for Computational Linguistics: ACL 2023*, pages 1538–1551, Toronto, Canada. Association for Computational Linguistics.